

EECS 841: Computer Vision

Lecture 7

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Outline

Binary Image Processing

- Morphological Operations
- Labeling Regions
- Computing Distance in a Digital Image
- Region Properties
- Skeletonization
- Boundary Representations

Morphology: Introduction and Uses

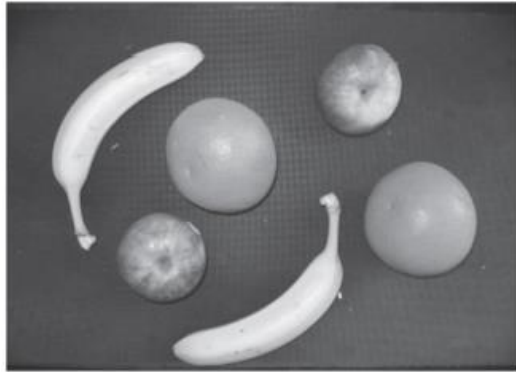
- **Mathematical morphology:** an entire branch of mathematics that has been developed to process images by considering the shape of the pixel regions.
 - Image pre-processing (noise filtering, shape simplification)
 - Enhancing object structures (skeletonization, thinning, convex hull, object marking)
 - Segmentation of the object from background
 - Quantitative descriptors of objects (area, perimeter, projection, Euler-Poincaré characteristics)

Morphology

Mathematical Morphology (cont.)

- The language of mathematical morphology is **set theory**.
- It is a mathematical tool for investigating **geometric structure** in an image.
- It is an approach for processing a digital image based on its **shape**.

Let's Begin!



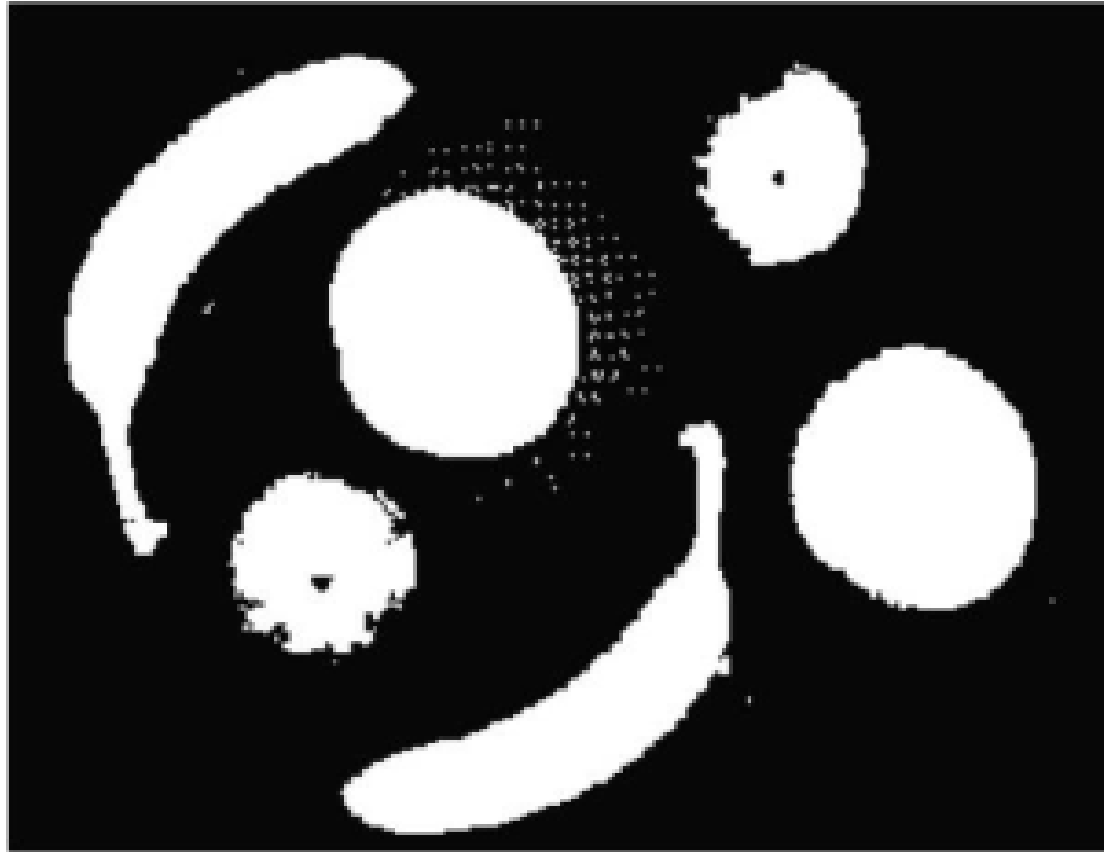
How can you binarize the image?

Ans:

Figure 4.1 Left: A grayscale image of several types of fruit on a dark conveyor belt. Right: A binary image resulting from thresholding. The white pixels are on and indicate the foreground, while the black pixels are off and indicate the background.



Think!



What would be the better version
of this binary image?

Think

Maybe this?



Sean Birchfield



Sean Birchfield

How to Achieve?

Ans: Erosion/ Dilation



Dilations make objects thicker, can be used to fill in holes, multiple applications can result in image which looks different from input



Erosions remove pixels from object boundaries, used to simplify structure of object. Objects with width one will disappear.

Erosion/ Dilation and Morphological Operation

- Thinning/ growing by the original shape
- Shape-based transformation = Morphological operation
- More precisely, Minkowski operation!

Morphology Goals

- To simplify image data, preserve essential shape characteristics, and eliminate noise
- Allows the underlying shape to be identified and reconstructed optimally from its noisy distorted forms.

Morphology: Main Idea

- Morphological operators often take a binary image and a structuring element as input and combine them using a set operator (intersection, union, inclusion, complement).
- The structuring element is shifted over the image and at each pixel of the image its elements are compared with the set of the underlying pixels.
- If the two sets of elements match the condition defined by the set operator (e.g. if set of pixels in the structuring element is a subset of the underlying image pixels), the pixel underneath the origin of the structuring element is set to a pre-defined value (0 or 1 for binary images).
- A morphological operator is therefore defined by its **structuring element** and the **applied set operator**.

Re-Introduction: Erosion/ Dilation

- Erosion/ Dilation = Thinning/ growing by the original shape
- Shape-based transformation = Morphological operation
- More precisely, Minkowski operation!

Thus, to understand Dilation/Erosion in CV, we should first learn about the Minkowski Addition/Subtraction.
Minkowski Addition/Subtraction are two special types of morphological operation applied to **BINARY IMAGES***!

Binary Image as a Set

- A **binary image** is an array of values such that $I(x, y)$ returns 1 or 0 for each pixel location (x, y) .
- Two fundamental set operators are *union* and *intersection*:

$$\mathcal{A} \cup \mathcal{B} \equiv \{z : z \in \mathcal{A} \text{ or } z \in \mathcal{B}\} \quad (\text{union})$$

$$\mathcal{A} \cap \mathcal{B} \equiv \{z : z \in \mathcal{A} \text{ and } z \in \mathcal{B}\} \quad (\text{intersection})$$

Minkowski Addition

1. Minkowski Addition ($\oplus$)

The Minkowski sum of two sets A and B in Euclidean space is defined as:

$$A \oplus B = \{a + b \mid a \in A, b \in B\}$$

Explanation:

For every point a in set A and every point b in set B , we add them together and include the result in the new set.

Relation between Minkowski Addition and Dilation:

1. Minkowski Addition ($A \oplus B$) $\rightarrow$ "A Grows by B"

- A is the main object (e.g., an image, shape, or region).
- B is the structuring element (the shape used to grow A).
- The result is a new set where every point in A is shifted by every point in B , effectively **expanding** A .

(Or vice-versa, i.e., B grows by A if A is the Structuring element)

Minkowski Subtraction

2. Minkowski Subtraction ($\ominus$)

The Minkowski difference of two sets A and B is given by:

$$A \ominus B = \{x \mid x + b \in A, \forall b \in B\}$$

Explanation:

We keep only those points x where adding **any** point from B would still keep the result inside A . This effectively "shrinks" A by the shape of B .

Relation between Minkowski Subtraction and Erosion:

2. Minkowski Subtraction ($A \ominus B$) $\rightarrow$ "A Shrinks by B"

- A is still the main object.
- B is the structuring element.
- The result removes parts of A where B doesn't fully fit, effectively **shrinking** A .

(Or vice-versa, i.e., B shrinks by A if A is the Structuring element)

Summary: Dilation and Erosion

- Minkowski addition grows (or “dilates”) the region by increasing its size.
- Minkowski subtraction shrinks (or “erodes”) the region by decreasing its size.

What Can be Two Common Usage:

- Noise removal
- Subject separation from background

Summarize Formally: Dilation and Erosion

- **Dilation:** The dilation of an image I with a structuring element B is:

$$I \oplus B = \bigcup_{b \in B} (I + b)$$

- ◆ This means shifting I by all points in B , causing it to **expand**.
- ◆ Equivalent to Minkowski **addition**.

- **Erosion:** The erosion of an image I with structuring element B is:

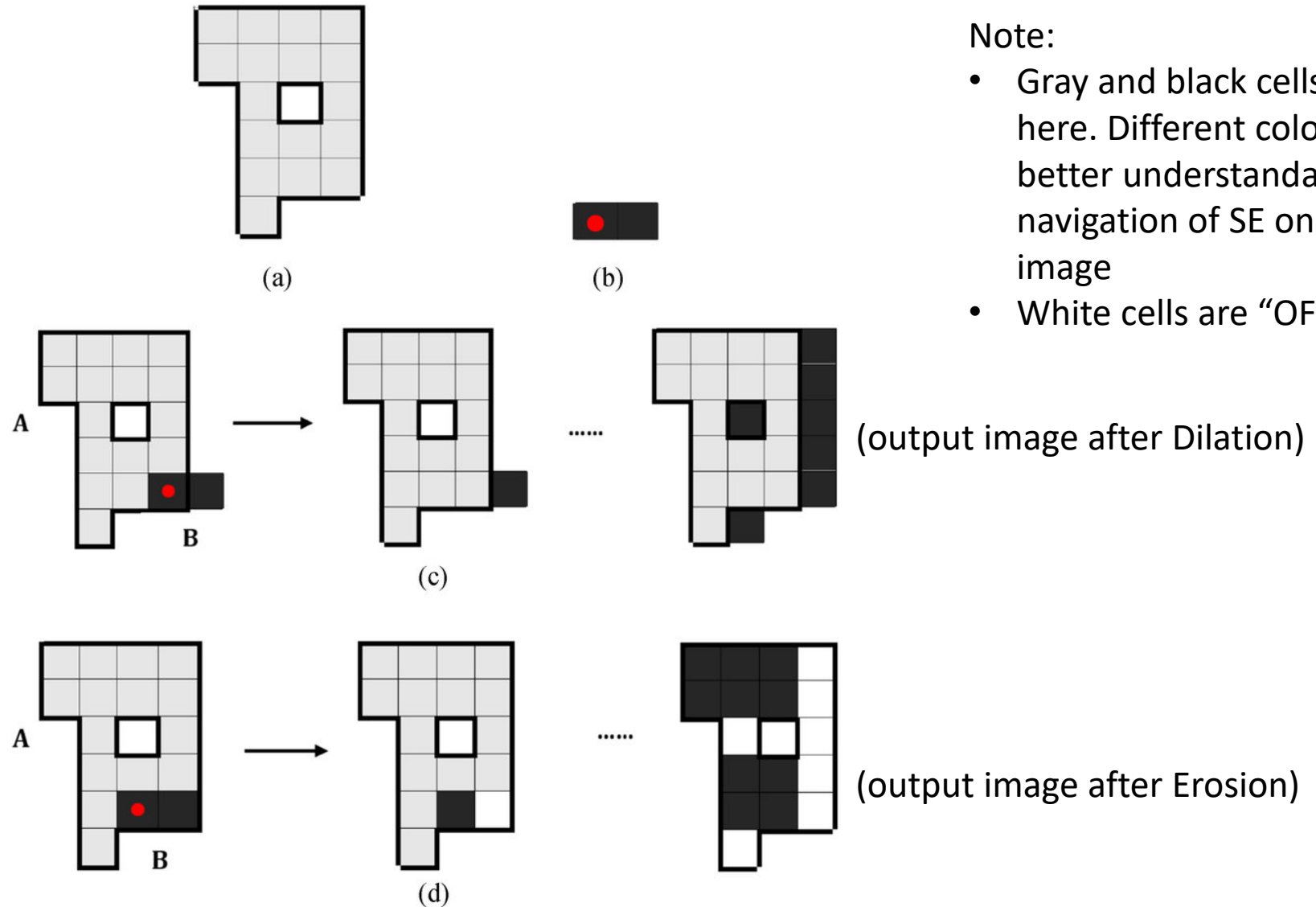
$$I \ominus B = \{x \mid B + x \subseteq I\}$$

- ◆ This means keeping only those points where the entire structuring element B fits inside I .

- ◆ Equivalent to Minkowski **subtraction**.

This means: x are the points in I . Whenever you shift x by B , it still fits inside I , only they are going to contribute towards your final image

Quick Demonstration



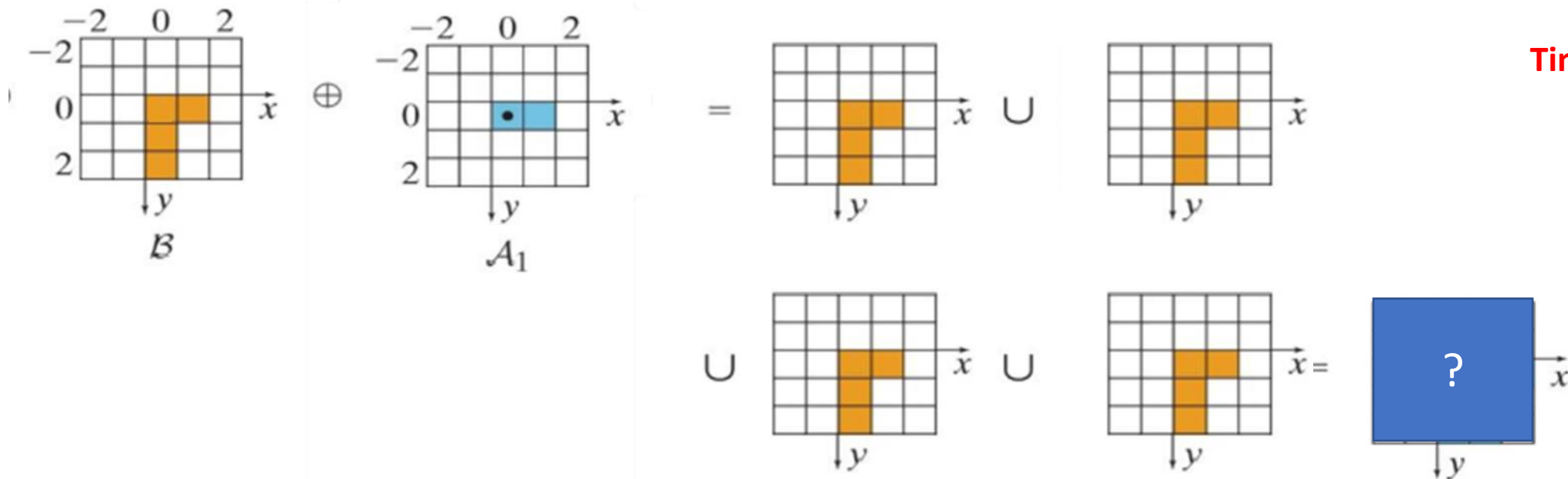
Note:

- Gray and black cells are “ON” here. Different colors are used for better understandability of the navigation of SE on the original image
- White cells are “OFF” here

Practice: Dilation of Original Image

Two sets: B (original image) and A (structuring element)

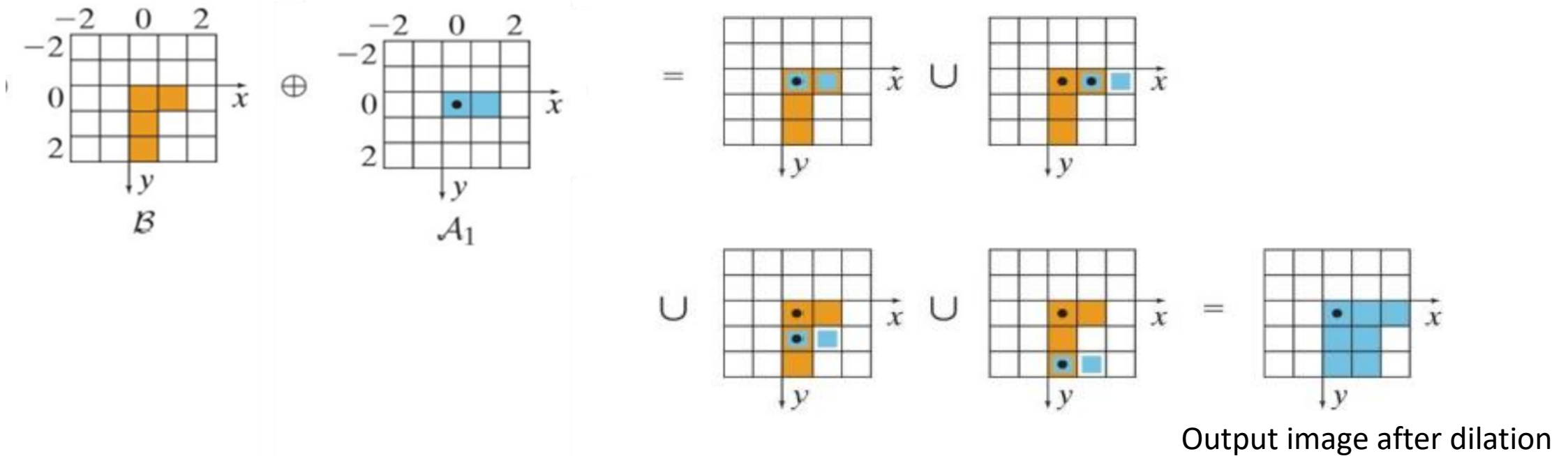
- Origin of SE counts
- Dilation process: overlay SE (=A's here) origin in all the pixels of original image (=B here) $\rightarrow$ if B has n ON pixels, then there will be n times overlaying $\rightarrow$ during each overlaying, we determine if the overlay is causing any new pixel to get ON (either by A or B or both) $\rightarrow$ final output image will be a collection of all the previous and newly ON-ed cells



Time: 3 min

Output image after dilation

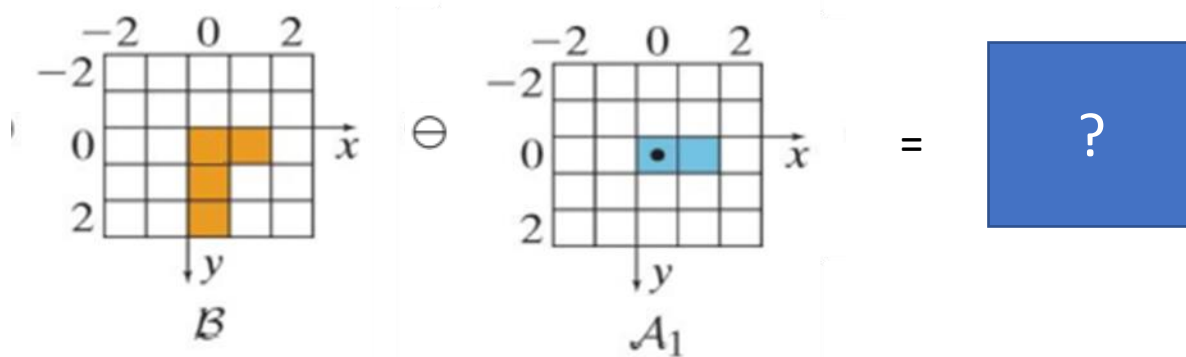
The Answer



- The original image “B” is translated by structuring element “A”
- The output is drawn by Minkowski Addition/union, which means that each ON cell in the output has occurred from at least one of the ON cells from the intermediate translations

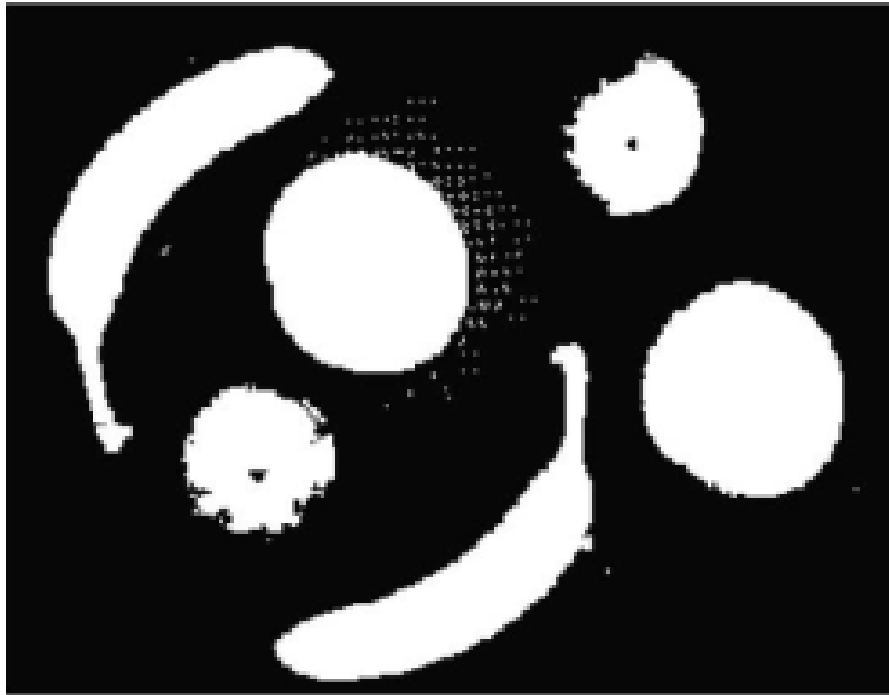
Practice: Erosion of Original Image

Two sets: B (original image) and A (structuring element)



Time: 2 min

Can you improve this by Erosion and/or Dilation



Summary: Morphological Operation in Image

- Morphological image processing deals with the processing of images based on the shape of the features in an image
- Useful during segmentation on binary images
- Structuring elements is a key factor in morphological operation since it probes the image under study
- The SE should have an origin
- The origin needs to be overlayed with all of the ON pixels in the input image → a decision is made based on the overlay → and finally the output image is obtained
- Three possible conditions are related to SE: fit, hit, and miss

Morphological Operation in Image

Fit:

11	0	0
11	0	1
11	1	1

When all cells under watch are matching with SE

Suppose: SE =

1
1
1

Hit:

11	0	0
10	0	1
10	1	1

When cells under watch are partially matching with SE

Miss:

10	0	0
10	0	1
10	1	1

When No cells under watch are partially matching with SE

Dilation and Erosion

Dilation enlarges the foreground.

Condition: assign “ON” cell to output if there is a “Fit” or “Hit” condition during morphological operation

Erosion enlarges the background or shrinks the foreground.

Condition: assign “ON” cell to output if there is a “Fit” condition during morphological operation

Dilation Example: step 1

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

1
1
1

origin

Question: Dilate A by SE

1					
10	0	0	0	0	0
10	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

Output after dilation:

0					

Dilation Example: step 1

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

1
1
1

← origin

1					
10	0	0	0	0	0
10	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

Output after dilation:

0					

Dilation Example: step 2

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

1
1
1

origin

	1				
0	1 0	0	0	0	0
0	1 0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

Output after dilation:

0	0				

Dilation Example: step 3

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

1
1
1

origin

		1			
0	0	10	0	0	0
0	0	11	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

CONTINUE
(time: 3 min)

Output after dilation:

0	0	1			

Dilation Example: Final result

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

1
1
1

← origin

Output after dilation:

0	0	1	1	0	0
0	1	1	1	1	0
0	1	1	1	1	0
0	1	1	1	1	0
0	0	1	1	0	0

Erosion Example: step 1

A

1	1	1	1	1	1
1	1	0	0	1	1
1	0	0	0	0	1
1	1	0	0	1	1
1	1	1	1	1	1

1
1
1

← origin

Question: Erode A by SE

1					
11	1	1	1	1	1
11	1	0	0	1	1
1	0	0	0	0	1
1	1	0	0	1	1
1	1	1	1	1	1

0					

Erosion Example: step 2

1	1	1	1	1	1
1	1	0	0	1	1
1	0	0	0	0	1
1	1	0	0	1	1
1	1	1	1	1	1

1
1
1

← origin

	1				
1	11	1	1	1	1
1	11	0	0	1	1
1	0	0	0	0	1
1	1	0	0	1	1
1	1	1	1	1	1

0	0				

Erosion Example: step 3

1	1	1	1	1	1
1	1	0	0	1	1
1	0	0	0	0	1
1	1	0	0	1	1
1	1	1	1	1	1

1
1
1

← origin

CONTINUE
(time: 3 min)

		1			
1	1	11	1	1	1
1	1	10	0	1	1
1	0	0	0	0	1
1	1	0	0	1	1
1	1	1	1	1	1

0	0	0			

Erosion Example: step 7

1	1	1	1	1	1
1	1	0	0	1	1
1	0	0	0	0	1
1	1	0	0	1	1
1	1	1	1	1	1

1
1
1

← origin

1 1	1	1	1	1	1
1 1	1	0	0	1	1
1 1	0	0	0	0	1
1	1	0	0	1	1
1	1	1	1	1	1

0	0	0	0	0	0
1					

Erosion Example: step 8

1	1	1	1	1	1
1	1	0	0	1	1
1	0	0	0	0	1
1	1	0	0	1	1
1	1	1	1	1	1

1
1
1

← origin

1	11	1	1	1	1
1	11	0	0	1	1
1	10	0	0	0	1
1	1	0	0	1	1
1	1	1	1	1	1

0	0	0	0	0	0
1	0				

Erosion Example: step 9

1	1	1	1	1	1
1	1	0	0	1	1
1	0	0	0	0	1
1	1	0	0	1	1
1	1	1	1	1	1

1
1
1

← origin

1	1	11	1	1	1
1	1	10	0	1	1
1	0	10	0	0	1
1	1	0	0	1	1
1	1	1	1	1	1

0	0	0	0	0	0
1	0	0			

Erosion Example: step 12

1	1	1	1	1	1
1	1	0	0	1	1
1	0	0	0	0	1
1	1	0	0	1	1
1	1	1	1	1	1

1
1
1

← origin

1	1	1	1	1	11
1	1	0	0	1	11
1	0	0	0	0	11
1	1	0	0	1	1
1	1	1	1	1	1

0	0	0	0	0	0
1	0	0	0	0	1

Erosion Example: random step

1	1	1	1	1	1
1	1	0	0	1	1
1	0	0	0	0	1
1	1	0	0	1	1
1	1	1	1	1	1

1
1
1

← origin

1	1	1	1	1	1
1	1	0	0	1	1
1	0	0	0	10	1
1	1	0	0	11	1
1	1	1	1	11	1

0	0	0	0	0	0
1	0	0	0	0	1
1	0	0	0	0	1
				0	

Erosion Example: random step

1	1	1	1	1	1
1	1	0	0	1	1
1	0	0	0	0	1
1	1	0	0	1	1
1	1	1	1	1	1

1
1
1

← origin

1	1	1	1	1	1
1	1	0	0	1	1
11	0	0	0	0	1
11	1	0	0	1	1
11	1	1	1	1	1

0	0	0	0	0	0
1	0	0	0	0	1
1	0	0	0	0	1
1				0	

Erosion Example: final output

1	1	1	1	1	1
1	1	0	0	1	1
1	0	0	0	0	1
1	1	0	0	1	1
1	1	1	1	1	1

1
1
1

← origin

After erosion:

0	0	0	0	0	0
1	0	0	0	0	1
1	0	0	0	0	1
1	0	0	0	0	1
0	0	0	0	0	0